

Preservation versus Selection in Koide's Charged-Lepton Mass Relation

A geometric reformulation and a test of S_3 -invariant flows

CosmicThinker25

Independent researcher

Technical and computational assistance: Claude (Anthropic)

Technical draft v0.4 — June 26, 2026

Abstract

Koide's charged-lepton relation, $K = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 \simeq 2/3$, is one of the most precise empirical regularities in the flavour sector. Its geometric reading (the 45° cone) and its scheme dependence (pole versus running masses) are known. In this note we make explicit a distinction that is often blurred: *preserving* the relation against radiative corrections versus *selecting* its value dynamically. Normalising the root masses on the unit sphere, $x_i = \sqrt{m_i}/\sqrt{\sum_j m_j}$ with $\sum_i x_i^2 = 1$, the relation reduces *exactly* to a condition on the second elementary symmetric invariant $e_2 = \sum_{i<j} x_i x_j$:

$$K = \frac{1}{1 + 2e_2}, \quad K = \frac{2}{3} \iff e_2 = \frac{1}{4}.$$

Koide's problem is thereby concentrated into a minimal question: why is $e_2 = 1/4$? We show that (i) the Standard-Model renormalisation-group flow crosses $K = 2/3$ without treating it as a fixed point, and (ii) within a minimal class of S_3 -invariant gradient flows on S_+^2 , generated by low-order potentials in (e_2, e_3) , the attractors concentrate on the democratic point ($e_2 = 1$) and the hierarchical vertices ($e_2 = 0$), while the Koide band is *underpopulated* relative to its geometric null. The result is deliberately **negative and bounded**: minimal S_3 symmetry does not select Koide within the studied class, and Sumino-type mechanisms preserve but do not select it. Any complete explanation must produce $e_2 = 1/4$ from an independent physical reason rather than insert it as input.

1 Introduction

Koide's relation [1, 2] ties the electron, muon and tau masses,

$$K = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} \simeq \frac{2}{3}, \tag{1}$$

to remarkable precision when physical (pole) masses are used. It has a simple geometric reading, discussed by Foot and others [3]: the vector $v = (\sqrt{m_e}, \sqrt{m_\mu}, \sqrt{m_\tau})$ makes a 45° angle with the democratic axis $(1, 1, 1)$. This structure separates Koide from a purely arithmetic coincidence.

There is, however, a well-established difficulty. The relation works with pole masses but degrades with running masses defined in $\overline{\text{MS}}$: Xing and Zhang [4] showed that, evaluated with running masses, the deviation from $2/3$ at M_Z is of order 10^{-3} , against $\sim 10^{-5}$ for pole masses. This suggests that, if Koide has physical content, it is not a simple relation among high-scale running Yukawas but is tied to infrared quantities or protected by some mechanism. Sumino's mechanism [5, 6] addresses exactly this: a family gauge symmetry whose self-energy cancels

the QED correction separating pole and running masses. Sumino is therefore a *preservation* mechanism.

Our contribution is methodological and rests on a distinction worth making explicit:

$$\text{preserving Koide} \neq \text{selecting Koide}. \quad (2)$$

Preserving means preventing an already-satisfied relation from deteriorating; selecting means making it emerge from generic conditions. Neither implies the other. On that basis we pose a testable question: *can a minimal S_3 -symmetric geometric dynamics select $K = 2/3$?* The answer we find, within the studied framework, is negative, and it moreover localises the obstruction: the value to be selected, $e_2 = 1/4$, is not a point singled out by the three-generation symmetry.

Nothing geometric (the 45° cone), the running-mass failure, or the Sumino mechanism is original here; we include them as context. What we add is the reduction to $e_2 = 1/4$ as an operational formulation of the selection problem, the observation that the SM running does not select it either, and a quantitative test of S_3 -invariant flows with its geometric null.

2 Geometry of root masses

Define the normalised roots

$$x_i = \frac{\sqrt{m_i}}{\sqrt{m_e + m_\mu + m_\tau}}, \quad i \in \{e, \mu, \tau\}, \quad \implies \quad x_e^2 + x_\mu^2 + x_\tau^2 = 1. \quad (3)$$

The triplet defines a point on the positive octant of the unit sphere S_+^2 . Since (1) is invariant under common rescalings $m_i \rightarrow \lambda m_i$, the absolute scale is irrelevant: only ratios matter. Identifying configurations related by generation permutations, the natural space is the fundamental chamber S_+^2/S_3 (a spherical triangle).

With $n = (1, 1, 1)/\sqrt{3}$ the democratic axis,

$$\cos \theta = x \cdot n = \frac{x_e + x_\mu + x_\tau}{\sqrt{3}}. \quad (4)$$

From (1), using $\sqrt{m_i} = x_i \sqrt{\sum_j m_j}$,

$$K = \frac{1}{(x_e + x_\mu + x_\tau)^2}, \quad (5)$$

so $K = 2/3$ is equivalent to $(x_e + x_\mu + x_\tau)^2 = 3/2$, hence to

$$\cos^2 \theta = \frac{(x_e + x_\mu + x_\tau)^2}{3} = \frac{1}{2}, \quad \theta = 45^\circ. \quad (6)$$

Koide is, geometrically, a 45° cone about the democratic axis.

3 Reformulation in symmetric invariants

With the elementary invariants $e_1 = \sum_i x_i$, $e_2 = \sum_{i < j} x_i x_j$, $e_3 = x_1 x_2 x_3$, and using $\sum_i x_i^2 = 1$,

$$e_1^2 = \sum_i x_i^2 + 2 \sum_{i < j} x_i x_j = 1 + 2e_2. \quad (7)$$

Since $K = 1/e_1^2$,

$$\boxed{K = \frac{1}{1 + 2e_2}} \implies K = \frac{2}{3} \iff 1 + 2e_2 = \frac{3}{2} \iff \boxed{e_2 = \frac{1}{4}}. \quad (8)$$

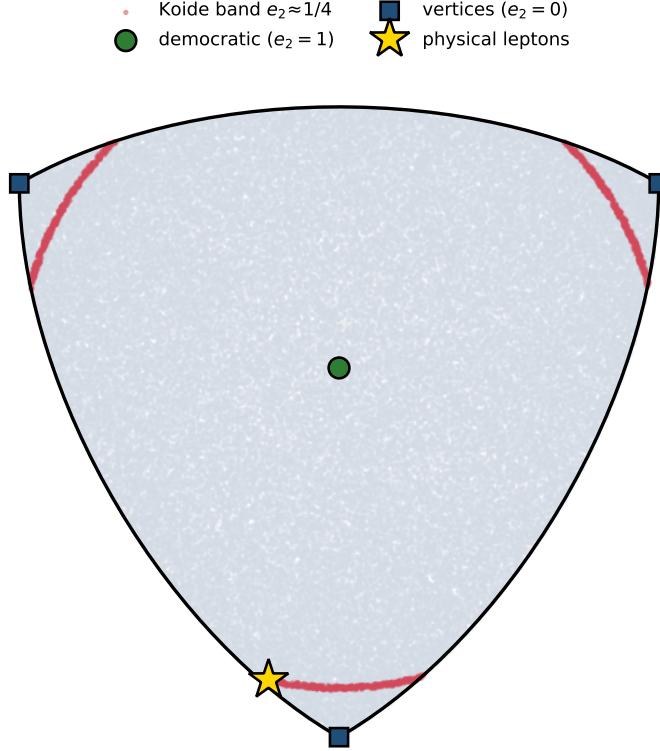


Figure 1: Fundamental chamber S_+^2/S_3 (spherical triangle). The democratic point ($e_2 = 1$) at the centre, the hierarchical vertices ($e_2 = 0$) at the corners, and the Koide band ($e_2 \approx 1/4$) as a latitude arc near the edges. The star marks the physical charged-lepton masses, which fall on the Koide band, away from the symmetry-privileged points.

The problem ceases to be a relation among three masses and becomes the origin of a single number: why the second symmetric invariant equals $1/4$. (Full derivation in Appendix A.)

With pole masses $m_e = 0.51100$, $m_\mu = 105.658$, $m_\tau = 1776.86$ MeV, the normalised roots are $x_{\text{phys}} \simeq (0.0165, 0.2369, 0.9714)$ and

$$e_2^{\text{phys}} = 0.25001, \quad (9)$$

indistinguishable from $1/4$ within the experimental uncertainty of m_τ (the fifth digit of e_2^{phys} is not significant). Koide's closeness is the extreme proximity of the physical point to the surface $e_2 = 1/4$.

4 Symmetry-privileged values

The S_3 symmetry distinguishes specific points of S_+^2 :

- **Democratic point** $x_1 = x_2 = x_3 = 1/\sqrt{3}$: here $e_2 = 1$, $K = 1/3$. Maximal symmetry among generations.
- **Hierarchical vertices** $(1, 0, 0)$ and permutations: here $e_2 = 0$, $K = 1$. Extreme hierarchy.

Koide requires $e_2 = 1/4$: neither the democratic maximum nor the hierarchical minimum, but an intermediate latitude. This is the core of the selection problem:

$$S_3 \text{ distinguishes } e_2 = 0 \text{ and } e_2 = 1, \text{ but does not evidently distinguish } e_2 = \frac{1}{4}. \quad (10)$$

5 The Standard-Model running does not select Koide either

Before geometry, it is worth ruling out the other natural flow of the problem: the renormalisation group. Were $K = 2/3$ a fixed point of the running, it would be selected dynamically without flavour symmetry. Integrating the SM charged-lepton Yukawa couplings at one loop and tracking $K(\mu)$ along the physical trajectory starting from $\overline{\text{MS}}$ masses at M_Z , one finds that K drifts monotonically, with $dK/d\ln\mu \sim +10^{-7}$ and no sign change over fourteen orders of magnitude in energy, and does not approach $2/3$. The RG crosses the neighbourhood of Koide’s condition without treating it as a fixed point.

The reason is structural. K is invariant under $m_i \rightarrow \lambda m_i$, and the leptonic running is almost entirely flavour-universal (the gauge terms and the Yukawa trace are common to the three generations and cancel in K). Only the self term $\propto y_i^2$ is non-universal, and it is tiny ($y_\tau \sim 10^{-2}$). Hence K barely moves and, crucially, does not have $2/3$ as an attractor. The one-loop integration (script `koide_running.py`) shows this explicitly:

μ	$K(\mu)$	$K - 2/3$	$dK/d\ln\mu$
M_Z	0.667929	$+1.26 \times 10^{-3}$	$+9.6 \times 10^{-8}$
10^3 GeV	0.667929	$+1.26 \times 10^{-3}$	$+9.9 \times 10^{-8}$
10^9 GeV	0.667930	$+1.26 \times 10^{-3}$	$+1.1 \times 10^{-7}$
10^{16} GeV	0.667932	$+1.27 \times 10^{-3}$	$+1.3 \times 10^{-7}$

(These values start from $\overline{\text{MS}}$ masses at M_Z , hence $K \neq 2/3$, consistent with the pole→running degradation noted in the introduction.) Along this physical trajectory K drifts monotonically and does not approach $2/3$ at any scale. Conclusion:

$$\text{neither the SM RG nor (as we show) minimal } S_3 \text{ geometry select } e_2 = 1/4. \quad (11)$$

6 Preservation versus selection

Sumino’s mechanism reads geometrically as follows: the QED correction, flavour non-universal, displaces the point off the surface $e_2 = 1/4$ when going from pole to running masses; the family gauge contribution cancels that displacement. Explicitly, the pole→ $\overline{\text{MS}}$ separation is a logarithm $\propto \frac{3}{2}(\alpha/\pi)\ln(M_Z/m_i)$ — the coefficient is the mass anomalous dimension — and undoing that leading logarithm removes the dominant displacement at first order and restores the relation to near-pole precision. Sumino *preserves*. But preserving is not selecting:

$$\text{preserving} = \text{keeping } e_2 = 1/4 \text{ against corrections}, \quad (12)$$

$$\text{selecting} = \text{making } e_2 = 1/4 \text{ emerge from generic conditions}. \quad (13)$$

The following sections ask whether a minimal S_3 -symmetric dynamics performs the second task.

7 S_3 -invariant gradient flows

On S_+^2 we consider potentials built from symmetric invariants. A minimal family is

$$V(e_2, e_3) = a e_2 + b e_2^2 + c e_3 + d e_3^2 + f e_2 e_3, \quad (14)$$

with $(a, b, c, d, f) \in \mathbb{R}$. *This ansatz is not the most general S_3 potential*: it is a low-order polynomial family, sufficient to probe whether the symmetry favours the Koide region. The gradient flow projected to the sphere is

$$\dot{x} = -\nabla_{S^2} V = -[\nabla_{\mathbb{R}^3} V - (\nabla_{\mathbb{R}^3} V \cdot x) x], \quad \nabla V = \frac{\partial V}{\partial e_2} \nabla e_2 + \frac{\partial V}{\partial e_3} \nabla e_3, \quad (15)$$

with $\nabla e_2 = (x_2 + x_3, x_1 + x_3, x_1 + x_2)$ and $\nabla e_3 = (x_2 x_3, x_1 x_3, x_1 x_2)$.

Remark 1 (Boundary treatment). *The positive octant has a boundary (some $x_i = 0$), where the vertices and edges live. In the integration we keep $x_i \geq 0$ by taking $|x_i|$ after each step and renormalising. This introduces a reflecting wall at $x_i = 0$: adequate to locate interior and boundary attractors, but it biases the rate of arrival at the vertices. We therefore report the frequencies of each type separately and do not interpret the vertex fraction as a physical probability. A cleaner integration would use internal coordinates of the spherical triangle (e.g. (e_2, e_3) as charts), avoiding the $\mathbb{R}^3$ projection; we leave this for future work.*

8 Exact analysis on the $x_1 = x_2$ branch

As an analytic laboratory we take the two-equal branch,

$$x_1 = x_2 = t, \quad x_3 = \sqrt{1 - 2t^2}, \quad e_2 = t^2 + 2t\sqrt{1 - 2t^2}. \quad (16)$$

This branch does *not* contain the physical point (which has three distinct components); it serves to see in a controlled way how Koide appears. Imposing $e_2 = 1/4$,

$$t^* = \sqrt{\frac{1}{4} - \frac{\sqrt{2}}{6}} \approx 0.1196, \quad x_3^* \approx 0.9856. \quad (17)$$

The democratic point ($t = 1/\sqrt{3}$) and the vertices are critical for *any* V by symmetry. For t^* to be critical of (14) one needs $dV/dt|_{t^*} = 0$, which imposes a non-trivial relation among the coefficients.

Remark 2 (What it shows and what it does not). *This is **not** a theorem over all S_3 potentials. It shows, on an analytically tractable branch and within the low-order polynomial ansatz in the invariants (e_2, e_3) , that: (i) the points the symmetry grants are the extremes $e_2 \in \{0, 1\}$; (ii) making Koide critical requires at least one relation among coefficients — it is not generic. With t^* fixed, “ V has a critical point at Koide on this branch” is a codimension- ≥ 1 condition in coefficient space; the full physical point (with its e_3) would impose more. The safe statement is: selecting $e_2 = 1/4$ requires a non-trivial relation among coefficients; it does not follow from S_3 alone.*

9 Numerical sweep

Geometric null. e_2 is not uniformly distributed on $[0, 1]$. Sampling points with the same measure as the sweep’s initial conditions (non-negative Gaussians in $\mathbb{R}^3$, normalised; 4×10^6 points) gives $\langle e_2 \rangle \approx 0.637$ and

$$P_{\text{geom}}(0.22 < e_2 < 0.28) \approx 2.84\%, \quad (18)$$

well below the 6% of a flat window. This 2.84% is the natural geometric null for the sampling measure used here; comparing with the flat-window 6% would be incorrect.

Protocol. For each potential, coefficients are sampled with a fixed seed, 8 initial conditions on S_+^2 are integrated ($dt = 0.02$, 1400–1500 steps), and the attractor is classified by its final e_2 : democratic ($e_2 > 0.95$), vertex ($e_2 < 0.05$), Koide band ($0.22 < e_2 < 0.28$) or other interior. We distinguish *per-trajectory* and *per-potential* frequency (fraction of potentials with at least one trajectory in the band).

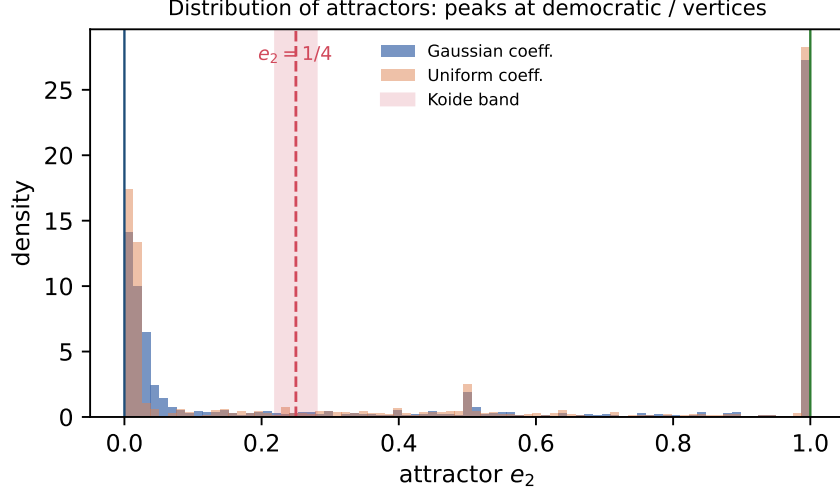


Figure 2: Distribution of attractors in e_2 for the two coefficient measures. The peaks sit at $e_2 = 0$ (vertices) and $e_2 = 1$ (democratic); the Koide band (shaded, $e_2 \approx 1/4$) is nearly empty.

Results. With the stated seeds and protocol, classifying with the reference band $0.22 < e_2 < 0.28$:

Attractor type	Gaussian coeff. (1500 $\times$ 8 traj.)	Uniform coeff. $[-1, 1]$ (1200 $\times$ 8 traj.)
Democratic ($e_2 > 0.95$)	45.7%	45.0%
Vertices ($e_2 < 0.05$)	34.5%	35.3%
Koide band ($0.22 < e_2 < 0.28$)	1.40%	1.51%
Other interior	18.5%	18.2%

Robustness across windows. The conclusion does not depend on the chosen band. The table below and Fig. 3 compare, for four windows, the Koide frequency (per-trajectory and per-potential) with its geometric null. In *all* windows and *both* measures, Koide stays below the null (ratio 0.4–0.6):

Window in e_2	Geom. null	Koide gauss. (traj./pot.)	Koide unif. (traj./pot.)
0.22–0.28	2.83%	1.40% / 1.40%	1.51% / 1.58%
0.23–0.27	1.88%	0.93% / 0.93%	1.09% / 1.17%
0.24–0.26	0.94%	0.41% / 0.47%	0.51% / 0.58%
0.245–0.255	0.47%	0.27% / 0.33%	0.18% / 0.25%

The per-potential frequency is very close to the per-trajectory one, indicating that Koide-reaching potentials are rare; we read this as rarity across potentials, not as evidence for small basins of attraction (indeed $f_{\text{pot}} \approx f_{\text{traj}}$ with $N_{\text{ini}} = 8$ implies that those few potentials send almost all their initial conditions to the band, not just one). In both measures and all four windows, Koide appears *below* its geometric null: the S_3 dynamics not only fails to select the Koide latitude but populates it less than geometric chance would, concentrating attractors at the symmetry fixed points.

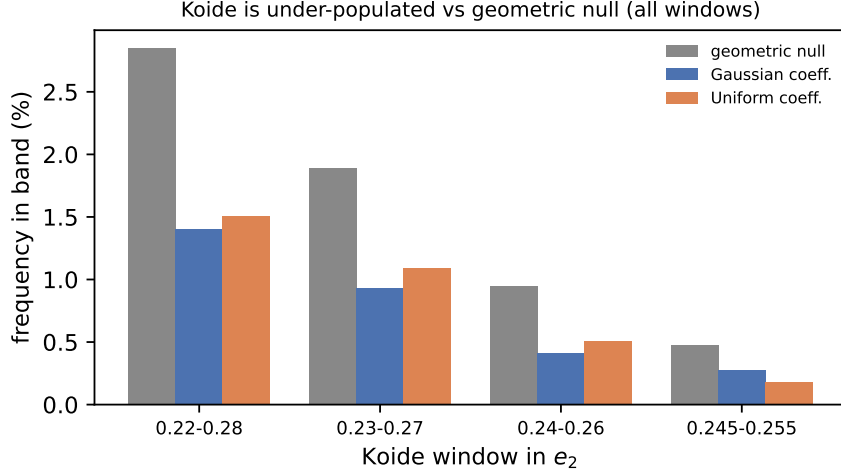


Figure 3: Frequency of attractors in the Koide band versus the geometric null, for four windows. In all of them, Koide is under-populated relative to geometric chance.

Reproducibility. All numbers and figures come from the scripts `koide_sweep.py` (sweep, explicit fixed seeds, NumPy) and `koide_running.py` (running of K , SciPy), attached as supplementary material; Appendix C summarises the procedure and includes the sweep core code.

Caveats. The result depends on the family (14), the measure over coefficients, the band definition, the discretisation and the boundary treatment. It is not an impossibility theorem.

10 Discussion and limitations

We separate four levels. *Empirical*: $K \simeq 2/3$ to 10^{-5} in pole masses, unexplained by the SM. *Geometric*: exactly equivalent to $e_2 = 1/4$. *Radiative*: Sumino preserves against QED. *Selection*: neither the RG (§5) nor minimal S_3 flows (§9) produce $e_2 = 1/4$ generically.

Limitations, explicitly: (i) the exact analysis is restricted to the $x_1 = x_2$ branch and the low-order polynomial ansatz in (e_2, e_3) ; (ii) not all critical points of every V_{S_3} are classified; (iii) frequencies depend on the measure over potentials; (iv) the boundary treatment biases the vertex rate; (v) richer mechanisms are not excluded (extended horizontal symmetries, non-polynomial terms, threshold effects, SUSY, specific representations). The defensible conclusion is

$$\text{minimal } S_3 \text{ symmetry does not select Koide within the studied class,} \quad (19)$$

and *not* “no theory can select Koide”.

11 Conclusions

Koide’s relation is reformulated exactly as $K = 1/(1 + 2e_2)$, so that $K = 2/3 \iff e_2 = 1/4$. The problem is not the 45° geometry itself but the origin of the value $1/4$. The points privileged by S_3 are the democratic one ($e_2 = 1$) and the vertices ($e_2 = 0$); Koide sits at an intermediate, undistinguished latitude. The SM running crosses $2/3$ without treating it as a fixed point, and low-order S_3 gradient flows leave it under-populated relative to its geometric null. Sumino-type mechanisms preserve the relation against radiative corrections but do not select its value.

Therefore any complete explanation must produce

$$e_2 = \frac{1}{4} \text{ from an independent physical reason, not as input.} \quad (20)$$

The minimal open question is whether there exists a flavour dynamics whose parameters land on the non-trivial relation placing the attractor at $e_2 = 1/4$ for an external reason — a charge, a dimension, a Casimir, a stability condition. Until then, Koide should be regarded as an empirical geometric anomaly: preservable in specific models, not selected by the minimal three-generation symmetry.

A Derivation of $K = 1/(1 + 2e_2)$

With $x_i = \sqrt{m_i}/\sqrt{\sum_j m_j}$ one has $\sum_i x_i^2 = 1$ and $\sqrt{m_i} = x_i \sqrt{\sum_j m_j}$, hence $\sum_i \sqrt{m_i} = \sqrt{\sum_j m_j} \sum_i x_i$ and

$$K = \frac{\sum_i m_i}{(\sqrt{\sum_j m_j} \sum_i x_i)^2} = \frac{1}{(\sum_i x_i)^2} = \frac{1}{\sum_i x_i^2 + 2 \sum_{i < j} x_i x_j} = \frac{1}{1 + 2e_2}.$$

Thus $K = 2/3 \iff e_2 = 1/4$.

B Koide point on the $x_1 = x_2$ branch

With $x_1 = x_2 = t$, $x_3 = \sqrt{1 - 2t^2}$, $e_2 = t^2 + 2t\sqrt{1 - 2t^2}$. Imposing $e_2 = 1/4$ gives $t^* = \sqrt{1/4 - \sqrt{2}/6} \approx 0.1196$, with $x_3^* \approx 0.9856$. It lies on the Koide surface but is not the physical point (which has three distinct components).

C Reproducible sweep scheme

1. Fix the seed. Sample coefficients (a, b, c, d, f) .
2. $V = ae_2 + be_2^2 + ce_3 + de_3^2 + fe_2e_3$.
3. Sample initial conditions on S_+^2 (non-negative Gaussians, normalised).
4. $\nabla V = (\partial_{e_2} V)\nabla e_2 + (\partial_{e_3} V)\nabla e_3$; project: $\nabla_{S^2} V = \nabla V - (\nabla V \cdot x)x$.
5. Integrate $\dot{x} = -\nabla_{S^2} V$; after each step $x \rightarrow |x|$, $x \rightarrow x/\|x\|$.
6. Classify by final e_2 . Compare with the null P_{geom} obtained with the *same* measure of initial conditions.

```
for _ in range(N_pot):
    a,b,c,d,f = sample_coeffs(rng)          # fixed seed
    x = abs(rng.normal(size=(N_ini,3))); x /= norm(x,axis=1,keepdims=True)
    for _ in range(N_steps):
        x = abs(x); x /= norm(x,axis=1,keepdims=True)
        e2 = x1*x2 + x1*x3 + x2*x3; e3 = x1*x2*x3
        de2 = stack([x2+x3, x1+x3, x1+x2]); de3 = stack([x2*x3, x1*x3, x1*x2])
        g = (a+2*b*e2+f*e3)*de2 + (c+2*d*e3+f*e2)*de3
        g = g - sum(g*x)*x                  # tangent projection
        x = x - dt*g
    classify_by_e2(x)                        # demo / vertex / koide / other
```


References

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